Topic 6 – Effects of technological developments

6.1 Effects of technological developments

Spec: Mass production – the consumer society, built-in obsolescence, the effect mass production has on employment; The 'new' industrial age of high-technology production – computers and the development and manufacture of products, miniaturisation of products and components, the use of smart materials, products from innovative applications; The global marketplace – multinational companies in developed and developing countries, manufacturing 'offshore' in developing countries and local and global production; Social, moral and ethical impacts

Key facts

• **Effects of mass production on employment?** Created many factory jobs and lowered skill requirements (de-skilling, repetitive tasks); automation now replaces many manual jobs but creates skilled technical jobs.
• **Three types of obsolescence?** Technical/functional (parts wear out or can't be repaired), style/aesthetic (goes out of fashion), technological/software (no longer supported or incompatible).
• **Benefits of built-in obsolescence to manufacturers?** Repeat sales and steady income, funds R&D, allows new technology releases, maintains market share.
• **Drawbacks of built-in obsolescence?** Waste and resource depletion, e-waste, consumer distrust and damaged reputation, right-to-repair legislation, cost to consumers.
• **How have computers changed product development?** CAD/virtual modelling and simulation, rapid prototyping, CAM/CNC, robotics, data integration (PDM/ERP), global collaboration, faster time to market.
• **What enabled smart watches?** Miniaturised integrated circuits and sensors, low-power processors, high-density batteries, OLED/LCD touchscreens, Bluetooth, inductive charging, thin alloy/composite cases.
• **Why manufacture offshore?** Lower labour and land costs, government incentives, access to raw materials and new markets, fewer regulations.
• **Ethical issues with offshoring?** Low pay, long hours, poor working conditions, child labour, weaker environmental and H&S laws, loss of jobs at home, transport emissions.
• **Benefits of local production?** Shorter lead times and transport, lower carbon footprint, easier quality control, supports local jobs, responsive to demand, 'Made in Britain' brand value.

Type	Example
Technical/functional – parts designed to wear out or not be repairable	Glued-in batteries, cheap bearings, printer chips
Style/aesthetic – goes out of fashion	Annual model changes, fast fashion, new colours
Technological/software – unsupported or incompatible	Phone OS updates ending, new charging standards

Offshore manufacture in developing countries	Local / home production
✔ Lower labour, land and energy costs; tax incentives	✔ Skilled workforce, high quality control
✔ Access to raw materials and growing markets	✔ Short supply chains, quick response, JIT friendly
✔ Brings jobs, investment and skills to the host country	✔ Lower transport emissions; "Made in Britain" value
✘ Long supply chains, transport costs and emissions, delays	✘ Higher wages and costs
✘ Exploitation risks: low pay, long hours, child labour, poor H&S	✘ Stricter regulations add cost
✘ Weaker environmental laws, reputational damage	
✘ Language, culture, political instability, exchange rates, tariffs	
✘ Loss of home jobs and skills	

Key terms

Mass production	Manufacturing very large quantities of standardised products, often on assembly lines with specialised machines and division of labour (e.g. Ford Model T, 1913).
Consumer society	A society where buying goods is central to people's lives and identity, encouraged by advertising, credit and cheap mass-produced products.
Built-in (planned) obsolescence	Deliberately designing products to have a limited life – technically (parts fail, non-replaceable batteries), stylistically (fashion) or through software support ending – to encourage repeat purchases.
Miniaturisation	Making components and products smaller, lighter and more powerful, driven by microelectronics (integrated circuits, MEMS sensors, surface-mount components).
Multinational company (MNC)	A company with operations in several countries, e.g. Apple, IKEA, Toyota.
Offshoring	Moving manufacturing to another country, often a developing country with lower labour costs.
Globalisation	The increasing interconnection of economies through trade, communication, transport and global supply chains.
Reshoring	Bringing manufacturing back to the home country (local production) – e.g. for shorter lead times, quality control or reputation.

Exam tip (2019 Q7a): In a location decision, consider cost, skills, infrastructure, distance to market, political stability, trade agreements/tariffs, supply chain, ethics, environment and reputation – and weigh them against the **level of automation**.